

ASSIGNMENT 3

Name: Ayush Bhargav

Roll No.: 2415001877

Q1 — Question

Task 1 – Project Workspace: Create the project workspace with src, docs, config and backup directories and library_info.txt. Demonstrate pwd, directory navigation, listing files, displaying file contents and directory structure.

```
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~
$ mkdir Ayush-GLA-LibrarySystem

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~
$ cd Ayush-GLA-LibrarySystem

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem
$ mkdir src docs config backup

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem
$ touch library_info.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem
$ nano library_info.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem
$ pwd
/c/Users/Ayush Bhargav/Ayush-GLA-LibrarySystem

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem
$ ls
backup/  config/  docs/  library_info.txt  src/

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem
$ cat library_info.txt
Name: Ayush Bhargav
Roll Number:2415001877
Subject: Application Build and Release Automation Lab
Project Name: Online Library Management System
Semester: 5th

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem
$ find .
.
./backup
./config
./docs
./library_info.txt
./src
```

ASSIGNMENT 3

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Q2 — Question

Task 2 – Git Repository: Initialize Git, configure username/email, check status, stage files, create the initial commit, modify library_info.txt, add the project objective and description, check changes, stage the modified file, create a second commit and display commit history.

```
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem
$ git init
Initialized empty Git repository in C:/Users/Ayush Bhargav/Ayush-GLA-LibrarySystem/.git/

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git config --global user.name "Ayush"

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git config --global user.email "ayushbhargav008@gmail.com"

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    library_info.txt

nothing added to commit but untracked files present (use "git add" to track)

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git add .
bash: git: command not found

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git add .
warning: in the working copy of 'library_info.txt', LF will be replaced by CRLF the next time Git touches it

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git commit -m "Initial project setup"
[master (root-commit) 22551a7] Initial project setup
 1 file changed, 5 insertions(+)
 create mode 100644 library_info.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ nano library_info.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git diff
warning: in the working copy of 'library_info.txt', LF will be replaced by CRLF the next time Git touches it
diff --git a/library_info.txt b/library_info.txt
index 739e85b..a12ace7 100644
--- a/library_info.txt
+++ b/library_info.txt
@@ -3,3 +3,11 @@ Roll Number:2415001877
 Subject: Application Build and Release Automation Lab
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 Semester: 5th
+
```

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Q2 — Question

Task 2 – Git Repository: Initialize Git, configure username/email, check status, stage files, create the initial commit, modify library_info.txt, add the project objective and description, check changes, stage the modified file, create a second commit and display commit history.

```
+
+
+
+Project Objective:
+To develop an online library management system for managing books and library-related operations efficiently.
+
+Project Description:
+The system will help manage library books and provide basic book management functionality.

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git add library_info.txt
warning: in the working copy of 'library_info.txt', LF will be replaced by CRLF the next time Git touches it

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git commit -m "Add project objective and description"
[master 5b2e714] Add project objective and description
1 file changed, 8 insertions(+)

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git log --online
fatal: unrecognized argument: --online

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$
```

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Q3 — Question

Task 3 – Feature Branch: Develop a Book Management Module on the book-feature branch. Create src/book.txt with a description and at least three features. Stage and commit the changes and demonstrate branch creation, switching, current branch, all branches and returning to main/master.

```
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git branch book-feature

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git switch book-feature
Switched to branch 'book-feature'

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (book-feature)
$ git branch
* book-feature
  master

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (book-feature)
$ nano src/book.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (book-feature)
$ git add src/book.txt
warning: in the working copy of 'src/book.txt', LF will be replaced by CRLF the next time Git touches it

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (book-feature)
$ git commit -m "Add Book Management Module"
[book-feature 7a2c8dc] Add Book Management Module
1 file changed, 10 insertions(+)
create mode 100644 src/book.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (book-feature)
$ git branch --show-current
book-feature

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (book-feature)
$ git branch
* book-feature
  master

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (book-feature)
$ git switch main
fatal: invalid reference: main

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (book-feature)
$
```

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Q4 — Question

Task 4 – Merge Conflict: Merge book-feature into main/master, create/use library-update, modify the same line in library_info.txt differently on the two branches, commit both changes, attempt the merge, identify and manually resolve the conflict, then stage, commit and display the final file, status and history.

```
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git switch master
git merge book-feature
Already on 'master'
Already up to date.

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git switch -c library-update
fatal: a branch named 'library-update' already exists

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ nano library_info.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git add library_info.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git commit -m "Update project name in library"
On branch master
nothing to commit, working tree clean

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git switch master
Already on 'master'

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ nano library_info.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git add library_info.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git commit -m "Modify project name in master"
On branch master
nothing to commit, working tree clean

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git merge library-update
Already up to date.

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ cat library_info.txt
Name: Ayush Bhargav
Roll Number: 2415001877
Subject: Application Build and Release Automation Lab
Project Name: Online Library Management System
Semester: 5th

Project Objective:
To develop an online library management system for managing books and library-related operations efficiently.
```

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Q4 — Question

Task 4 – Merge Conflict: Merge book-feature into main/master, create/use library-update, modify the same line in library_info.txt differently on the two branches, commit both changes, attempt the merge, identify and manually resolve the conflict, then stage, commit and display the final file, status and history.

```
On branch master
nothing to commit, working tree clean

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git merge library-update
Already up to date.

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ cat library_info.txt
Name: Ayush Bhargav
Roll Number: 2415001877
Subject: Application Build and Release Automation Lab
Project Name: Online Library Management System
Semester: 5th

Project Objective:
To develop an online library management system for managing books and library-related operations efficiently.

Project Description:
The system will help manage library books and provide basic book management functionality.

Project Name: GLA Digital Library System

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git status
On branch master
nothing to commit, working tree clean

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git log --oneline --graph --all
* c215eec (HEAD -> master) Finalize conflict resolution
* b02a695 Clean final merge resolution
*   ac43de8 Resolve merge conflict
| \
| * 4c4646a (library-update) Update project name in library
* | 9704312 Modify project name in master
| /
* 7a2c8dc (book-feature) Add Book Management Module
* 5b2e714 Add project objective and description
* 22551a7 Initial project setup

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ |
```

ASSIGNMENT 3

Name: Ayush Bhargav

Roll No.: 2415001877

Q5 — Question

Task 5 – Submission: Submit screenshots of important commands, the project directory, Git history, branch information, merge demonstration, and conflict and resolution demonstration. Students must be able to explain the commands used.

```
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~  
$ ls  
-1.14-windows.xml  
AppData/  
'Application Data'@  
Ayush-GLA-CourseRegistration/  
Ayush-GLA-HealthSystem/  
Ayush-GLA-LibrarySystem/  
Contacts/  
Cookies@  
CrossDevice/  
Desktop/  
Documents/  
Downloads/  
Favorites/  
IdeaProjects/  
IdeaSnapshots/  
'IntelliJ IDEA 2025.2.1'/  
Links/  
'Local Settings'@  
Music/  
'My Documents'@  
Myproject/  
NTUSER.DAT  
NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaal}.TM.blf  
NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaal}.TM.blf.cnpf  
NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaal}.TMContainer00000000000000000001.regtrans-ms  
NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaal}.TMContainer00000000000000000001.regtrans-ms.cnpf  
NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaal}.TMContainer00000000000000000002.regtrans-ms  
NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaal}.TMContainer00000000000000000002.regtrans-ms.cnpf  
NetHood@  
OneDrive/  
PrintHood@  
Recent@  
'Saved Games'/  
Searches/  
SendTo@  
'Start Menu'@  
TASK  
Templates@  
Videos/  
ashu/  
ashu-GLA-CourseRegistration/  
ayush/  
main.py  
ntuser.dat.LOG1  
ntuser.dat.LOG2  
ntuser.ini  
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~  
$ cat library_info.txt
```

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```
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ cat library_info.txt
Name: Ayush Bhargav
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Project Description:
The system will help manage library books and provide basic book management functionality.

Project Name: GLA Digital Library System

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ cd Ayush-GLA-LibrarySystem
bash: cd: Ayush-GLA-LibrarySystem: No such file or directory

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ Bhargav@LAPTOP-PK65VJLN
bash: Bhargav@LAPTOP-PK65VJLN: command not found

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ cd Ayush-GLA-LibrarySystem
cat library_info.txt
bash: cd: Ayush-GLA-LibrarySystem: No such file or directory
Name: Ayush Bhargav
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Project Objective:
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Project Name: GLA Digital Library System

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git branch
  book-feature
  library-update
* master

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git log --oneline --graph --all
```

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```
Project Name: GLA Digital Library System

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git branch
  book-feature
  library-update
* master

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ git log --oneline --graph --all
* c215eec (HEAD -> master) Finalize conflict resolution
* b02a695 Clean final merge resolution
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* 5b2e714 Add project objective and description
* 22551a7 Initial project setup

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$ cat src/book.txt
Book Management Module

Description:
The Book Management Module is used to manage books in the online library system.

Features:
1. Add new books
2. Search books
3. Update book information
4. Delete books

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-LibrarySystem (master)
$
```